

Conor O'Neil

Engineering Portfolio

Principal Mechanical Engineer | Product Development | Robotics and Automation

With over 12 years of developing and delivering complex electromechanical systems from concept through production, I'm ready to handle all types of engineering problems.

I have a deep passion for engineering – I've been designing and building projects my entire life!

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Professional work shown is limited to publicly released or non-proprietary material

Scoop

Robotic Trailer Unloader

ROLE & SCOPE

Technical lead and hands-on hardware engineer from early prototype through commercial deployment

SYSTEM ARCHITECTURE

Defined the product from the ground up – conceptualizing major components and shaping hardware and software functionality

END-EFFECTOR DESIGN

Prototyped package-removal concepts, grasp detection, and vacuum technologies

ARTICULATING CONVEYORS

Developed compact multi-axis conveyor and proprietary high-torque drivetrain systems

COMMERCIAL DEPLOYMENT

Deployed with multiple Fortune 50 customers



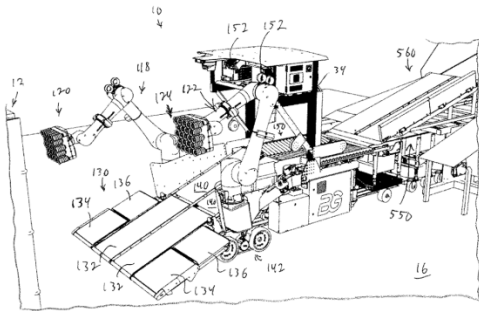
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System Architecture & Package Flow

ENGINEERING PROBLEM

- Interface a full-width robotic unloading system with narrower downstream conveyors
- Induct heavy packages from the floor without requiring a separate lifting mechanism



CONVEYOR LAYOUT

Devised a compact conveyor system to funnel packages from the full trailer width into the narrower takeaway path

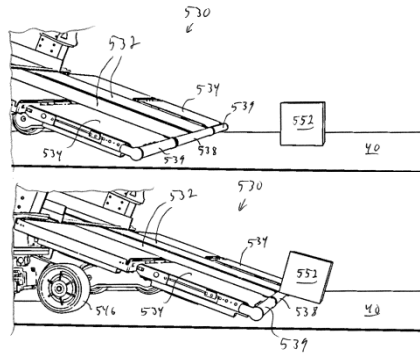


FIG. 26B

PACKAGE SCOOPING

Designed a high-torque nose-roller drivetrain and custom wheel system to lift heavy packages as the system advances

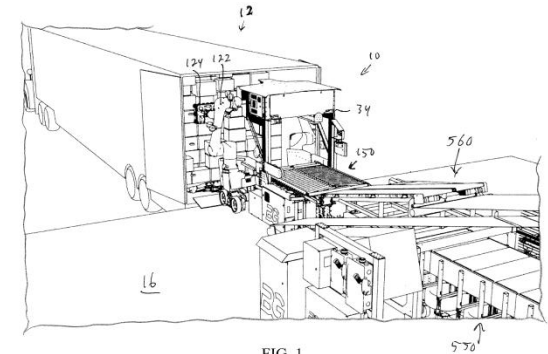


FIG. 1

SYSTEM PACKAGING

Architected the chassis and conveyor arrangement to fit the robotic unloading system within the constrained dock area

Suction Gripper End Effector

ENGINEERING PROBLEM

- Grasp multiple packages in an unstructured trailer environment
- Maintain vacuum grip when only a portion of the suction array contacts packages

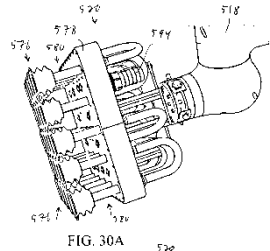


FIG. 30A

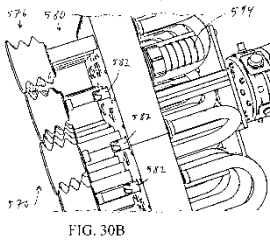


FIG. 30B

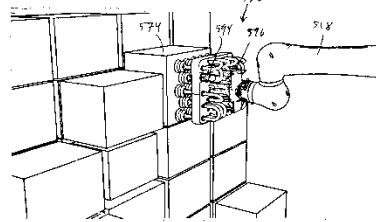


FIG. 36A

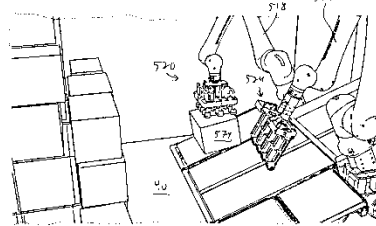


FIG. 36B

COMPLIANT GRIPPER DESIGN

Incorporated compliance mechanisms to conform the suction array to irregular, non-coplanar package surfaces

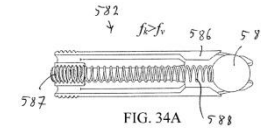


FIG. 34A

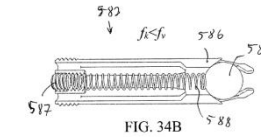


FIG. 34B

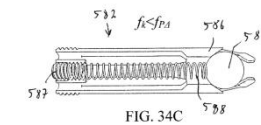


FIG. 34C

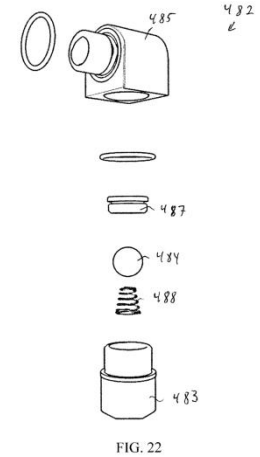


FIG. 22

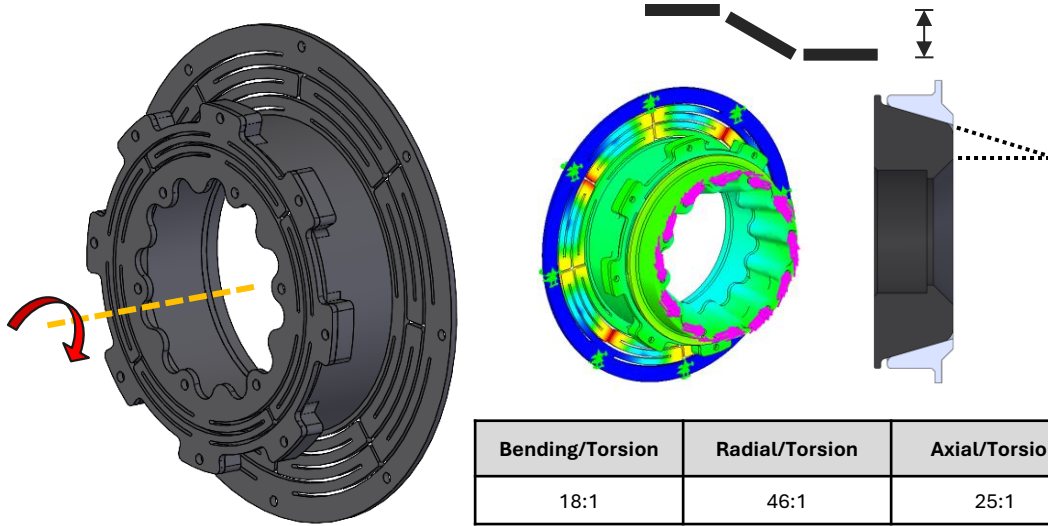
VACUUM CONTROL

Invented several high-flow vacuum-control concepts to automatically isolate unused suction cups and maintain vacuum at engaged cups

5-DoF Lens Focus System

ENGINEERING PROBLEM

- Position the lens relative to the imager precisely at the required focal distance
- Minimize hysteresis and parasitic motion to achieve $<1\ \mu\text{m}$ focus-positioning repeatability



$>10:1$ targeted for effective flexure design

CONCEPT GENERATION AND SIMULATION

Utilized two connected diaphragm sheet flexures to provide a total of 5 DoF. Simulated the flexure's stiffness to quantify resistance to undesired motion

DESIGN COMPATIBILITY

Developed a multi-lobed, angled lens interface to transmit torque while accommodating the range of production lens geometries

SYSTEM INTEGRATION

Integrated the flexure with a motorized drivetrain, adjustable focus-target stage, and automated focusing software to reliably position the lens

Imager Alignment Tolerance Analysis

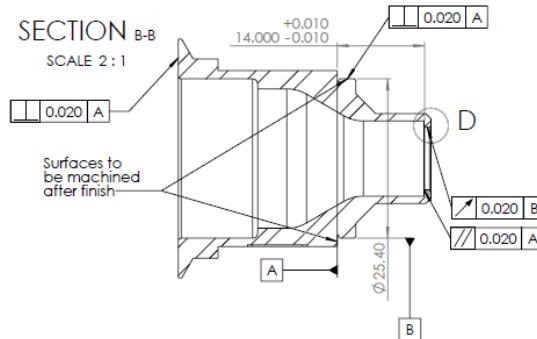
ENGINEERING PROBLEM

- Develop multiple automated machines capable of aligning the imager to the lens to within ± 0.15 mm
- Validate machine performance against the alignment specification

	Spec	Value	Units
	XY Alignment	148.35	um, +/-

	Uncertainty Source	Value (+/-)	Units	Type	Distribution	Standard Uncertainty Estimate	Standard Uncertainty (um)	Source of Value
Collet	Collet concentricity error	10.00	um	B	Rectangular	5.77	5.77	Tolerance on drawing
	Thread perpendicularity to C-mount flange	10.00	um	B	Rectangular	5.77	5.03	Tolerance on drawing
Stage	Yaw/Pitch error (whichever is bigger)	0.01	degrees	B	Rectangular	0.01	0.44	Stage specification
	Roll error	0.01	degrees	B	Rectangular	0.01	0.44	Stage specification
Stage Alignment	Straightness/Flatness error (whichever is bigger)	5.00	um	B	Rectangular	2.89	0.06	Stage specification
	Straightness/Flatness Alignment (whichever is bigger)	100.00	um	B	Rectangular	57.74	0.00	Estimate of stage alignment
Software Vision Job	Repeatability error of finding the pixel grid	3.00	pixels	A	Normal	1.73	10.00	Need to confirm with measurements
	Repeatability error of finding the C-mount	3.00	pixels	A	Normal	1.73	10.00	Need to confirm with measurements
CMM Ground Truth	CMM measured value offset	5.00	um	B	Rectangular	2.89	2.89	Systematic bias of CMM's measured values versus experiment
	CMM value fixturing error	2.00	um	A	Normal	1.15	1.15	Need to confirm with measurements
	CMM value repeatability error	2.00	um	A	Normal	1.15	1.15	Need to confirm with measurements
	CMM machine tolerance	2.08	um	B	Normal	1.20	1.20	Taken from the Hexagon CMM datasheet (2.0 + L/250)

Combined Uncertainty (RSS)	15.98	um, +/-
Expansion Factor (k)	3	99.7%
Expanded Uncertainty	47.94	um, +/-
Factor of Safety	3.09	



TOLERANCE DEVELOPMENT

Identified potential sources of variation and quantified their sensitivity to the overall alignment specification

GD&T AND CALIBRATION

Applied GD&T to critical components and documented calibration procedures with dial indicators and other metrology equipment

PRODUCTION VALIDATION

Created a CMM program and a statistical sampling plan to verify the machine's performance

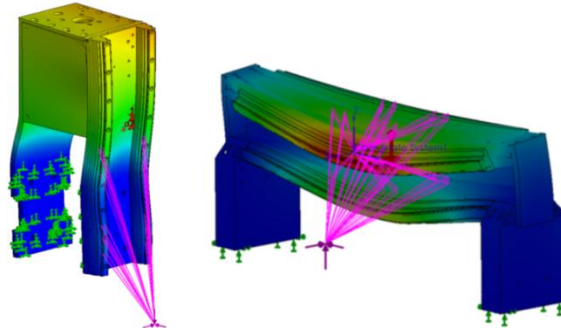
Enclosed CNC Machine

A transportable 3-axis gantry CNC designed, fabricated, and calibrated in a third-floor apartment



ELECTRICAL AND ENCLOSURE

Dual-wall MDF enclosure designed to reduce vibration and noise, with integrated servo controllers, Linux PC, I/O modules, and other electrical components



STRUCTURAL DESIGN

<25 μm

Target deflection under cutting load simulated using FEA